PIMPRI CHINCHWAD EDUCATION TRUST's.

PIMPRI CHINCHWAD COLLEGE OF ENGINEERING

(An Autonomous Institute)

T.Y. B. TECH	**Year:** 2025 – 26	**Semester:** II
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Department: Computer Engineering		**Division:** B
Course: Full Stack Development Lab		**Course Code:** BCE26VS01

Assignment 6

Assignment Title: Create Online appointment booking application / E-commerce portal for used items sales like car, bike etc. or similar type application using Node.js and Express use any NoSQL data set like MongoDB if required.

Objectives:

- To develop a responsive and user-friendly web application for scheduling medical consultations.
- To implement a secure Admin Dashboard for managing patient records and viewing appointment logs.
- To integrate frontend with backend using Fetch API
- To store appointment data in MongoDB database

Outcomes:

- Successfully created a portal where patients can book appointments without needing to register or log in.
- Built a protected Admin area that requires specific credentials to view sensitive patient data.
- Achieved seamless communication between the frontend and backend (Node.js/Mongoose).
- Gained practical experience in handling environment variables (.env) and local database administration using **MongoDB Compass**.

Theory:

Working Principle (How it Works):
The system follows a simple 3-step process:

1. **Booking:** A patient fills the form and clicks "Book Now." This sends the data to my server.
2. **Saving:** The server receives the info and saves it into a local database
3. **Viewing:** When the Admin logs in with the correct password, the server pulls all that saved info from the database and shows it in a table.

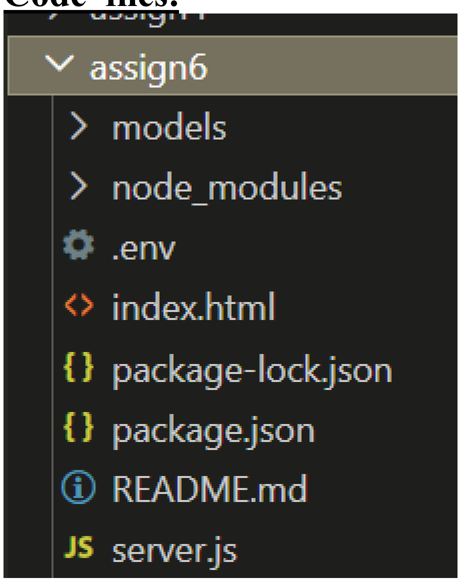
Technology Stack :

- Frontend : Used HTML, CSS, and Bootstrap 5 to make the website look professional and work on both mobile and laptops.
- Backend : Used Node.js and Express to handle the logic, like taking the form data and sending it to the database.
- Database : Used MongoDB (Local Server) to store all the appointment details safely.
- Library: Used Mongoose to help the code conect to MongoDB easily using simple JavaScript.
- Tools: Used VS Code for coding and MongoDB Compass to see the data inside my database visually.

Data Structure :
Since it’s a booking app, the "dataset" is just the info patients type in. Database saves these five things:

1. **Name:** Who is booking.
2. **Phone:** How to call them.
3. **Email:** How to reach them online.
4. **Date:** When they want the appointment.
5. **Status:** Automatically set to "Pending" until the admin checks it.

Code files:



MediBook

Admin Portal

Expert Medical Consultation

Book a specific slot with our available specialists at MediBook. No registration required.

Book Appointment

Easy Booking

Fill out our simple form and get instant confirmation.

Expert Doctors

Choose from our qualified specialists in various fields.

Secure & Private

Your information is safely stored and protected.

MediBook Admin

Logout

Master Appointment List

Patient	Doctor	Schedule	Phone	Status	Actions
Tanisha Mukesh Pande	Dr. Smith (Cardiologist)	2026-04-30 | 10:00 AM	09359166392	Confirmed	Delete
prachi	Dr. Patel (Dermatologist)	2026-02-02 | 10:00 AM	123456789	Confirmed	Delete

Appointment Confirmed!

Booked with Dr. Patel (Dermatologist) at 10:00 AM on 2026-02-02.

Go Back to Home

26°C Clear Search ENG IN 01:20:31 15-04-2026

Conclusion

This assignment successfully demonstrates a functional Full Stack Appointment Booking System. By integrating a responsive Bootstrap frontend with a Node.js and Express backend, I created a seamless flow for both patients and administrators.

The system efficiently handles data persistence using MongoDB, providing a reliable way to store and retrieve patient records. Overall, this project serves as a practical example of how modern web technologies can be used to build organized and professional medical scheduling applications.